

>120,000

confirmed COVID-19 deaths
(2020–2024)

255

deaths per 100,000 population

66%

fewer international tourist arrivals
in 2020

16.5%

peak unemployment rate (Q3
2020)

43%

peak test-positivity, far above
WHO 5% threshold

Spain was one of the European countries hit earliest and hardest by SARS-CoV-2. Madrid recorded community transmission already in late February 2020, and the national State of Alarm (14 March 2020) imposed one of the strictest lockdowns on the continent. With a tourism-dependent economy (~12% of GDP) and an ageing population (median age 44), Spain combines an unusually severe epidemiological shock with an unusually severe *economic* shock — which is why we chose it. Throughout the poster we benchmark Spain against **Poland**, a country with a similar population size but a very different pandemic timeline, demographic profile and policy response.

Definitions used. *Cases* = laboratory-confirmed SARS-CoV-2 infections reported by ISCH / OWID. *Excess mortality* = observed deaths minus the 2015–2019 baseline for the same calendar week (Eurostat). *Stringency index* = 0–100 score of containment policies (Oxford OxCGR). *CFR* = case-fatality rate = deaths / confirmed cases (a biased estimator, see Wave Analysis). All population-adjusted figures use 2020 INE / Eurostat populations.

Daily COVID-19 Cases (7-day smoothed): Spain vs F

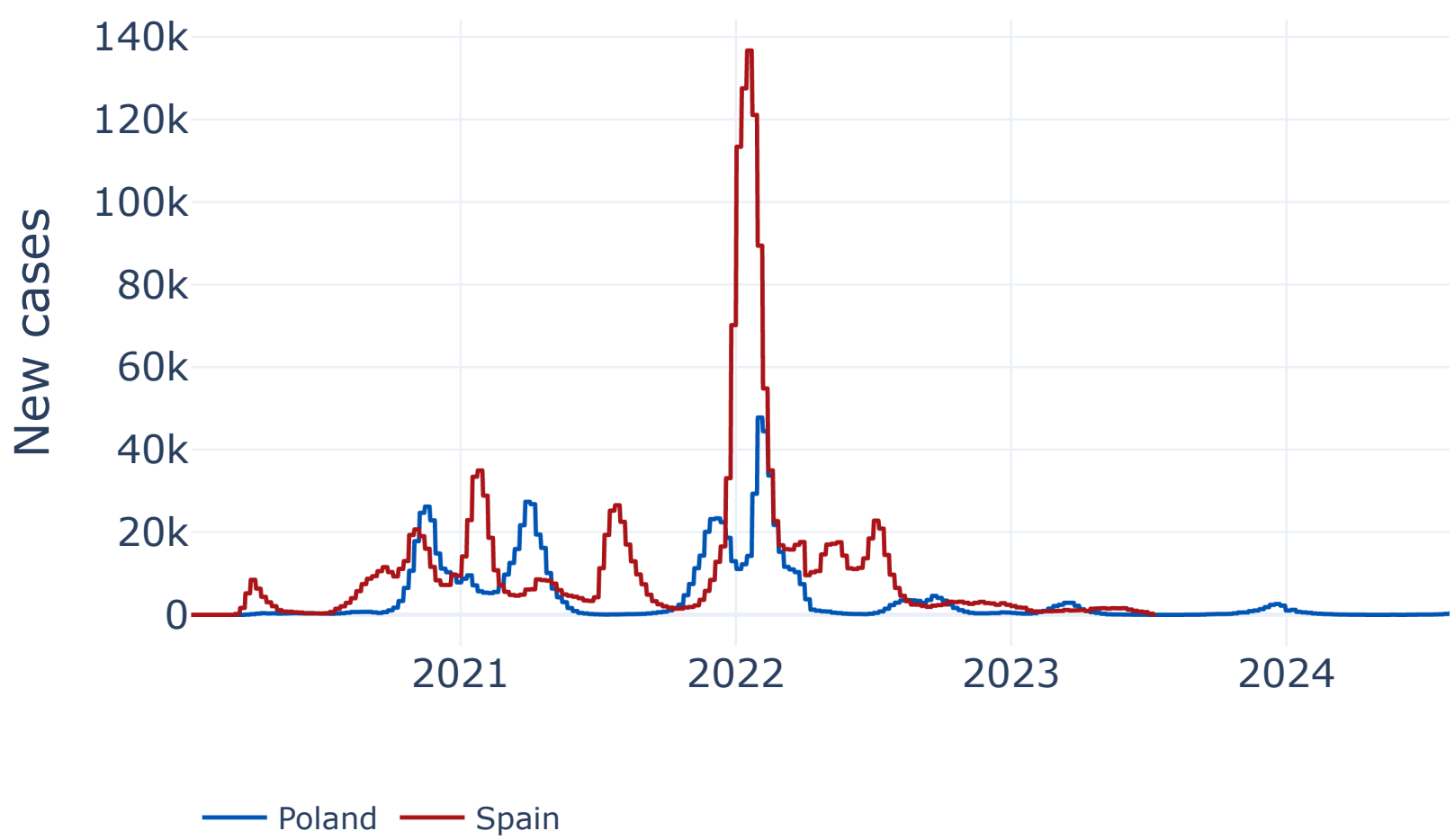


Fig. 1: Daily new cases per 100k inhabitants, Spain vs Poland. Five clear waves: the catastrophic first wave (Mar–May 2020, mostly invisible in confirmed cases because testing was unavailable), the summer 2020 resurgence, alpha (winter 20/21), delta (summer 21) and omicron (winter 21/22). After 2022, reporting changes flatten the curve.

The first wave dwarfs everything else in mortality but is barely visible in *confirmed* cases — a measurement artefact, not a real low. Excess mortality (next block) corrects for this and tells the real story.

Spain: Excess Mortality vs Reported COVID-19 Death

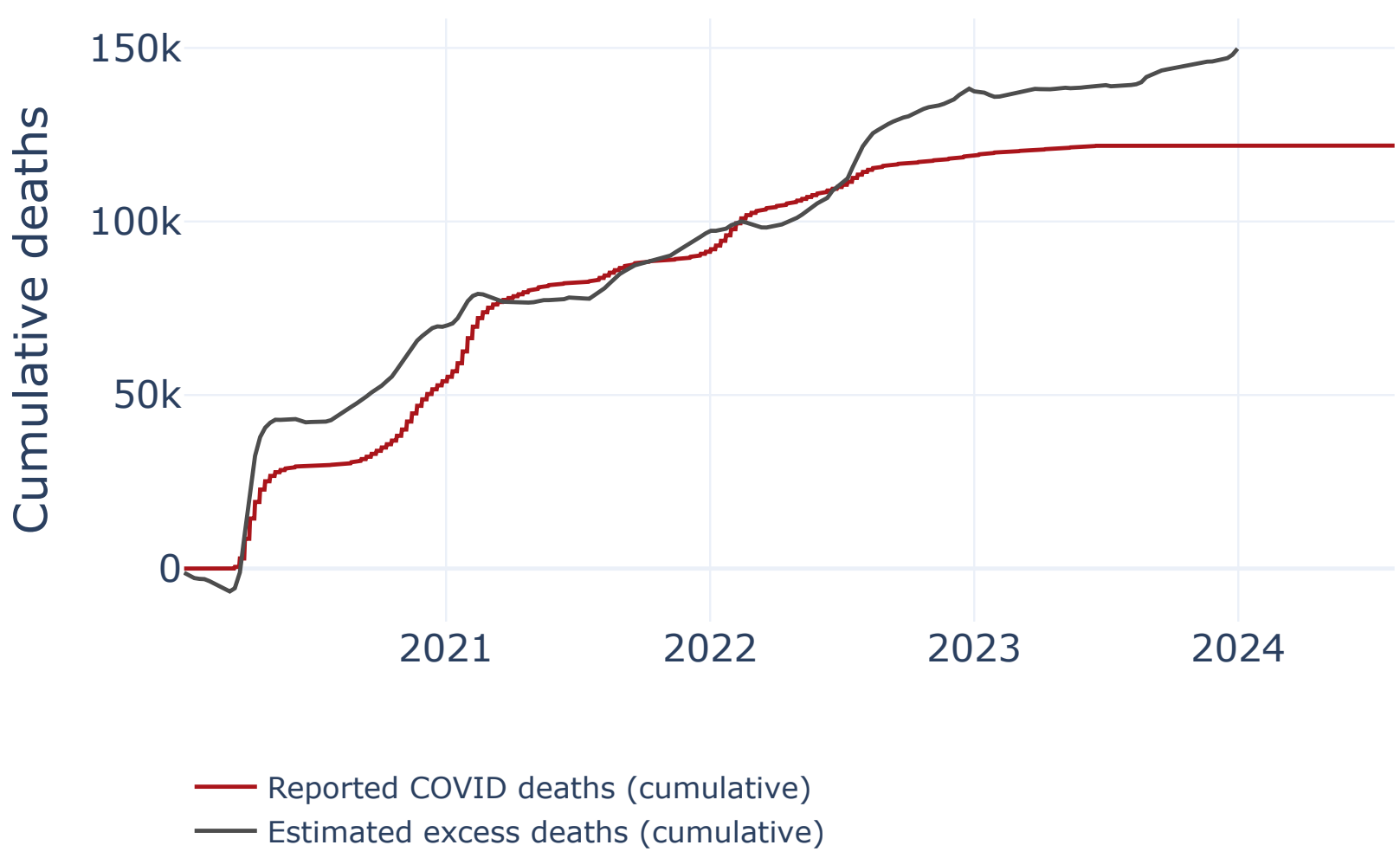


Fig. 2: Cumulative estimated excess deaths versus officially reported COVID-19 deaths in Spain (Our World in Data). The persistent gap — excess deaths running well above the official count — reveals substantial under-reporting, especially during the spring-2020 first wave.

Why this matters. Confirmed-death counts depend on testing capacity and definitions. Excess mortality does not — making it the most reliable single indicator of pandemic severity.

Spain Regional Case Burden by Pandemic Wave



Fig. 3: Cumulative cases per 100k by wave across Spain’s 17 autonomous communities. Madrid leads the first wave; later waves spread more uniformly, with inland communities (Aragón, Castilla y León) bearing higher per-capita burden than the islands.

Spain COVID-19 Deaths by Age and Sex

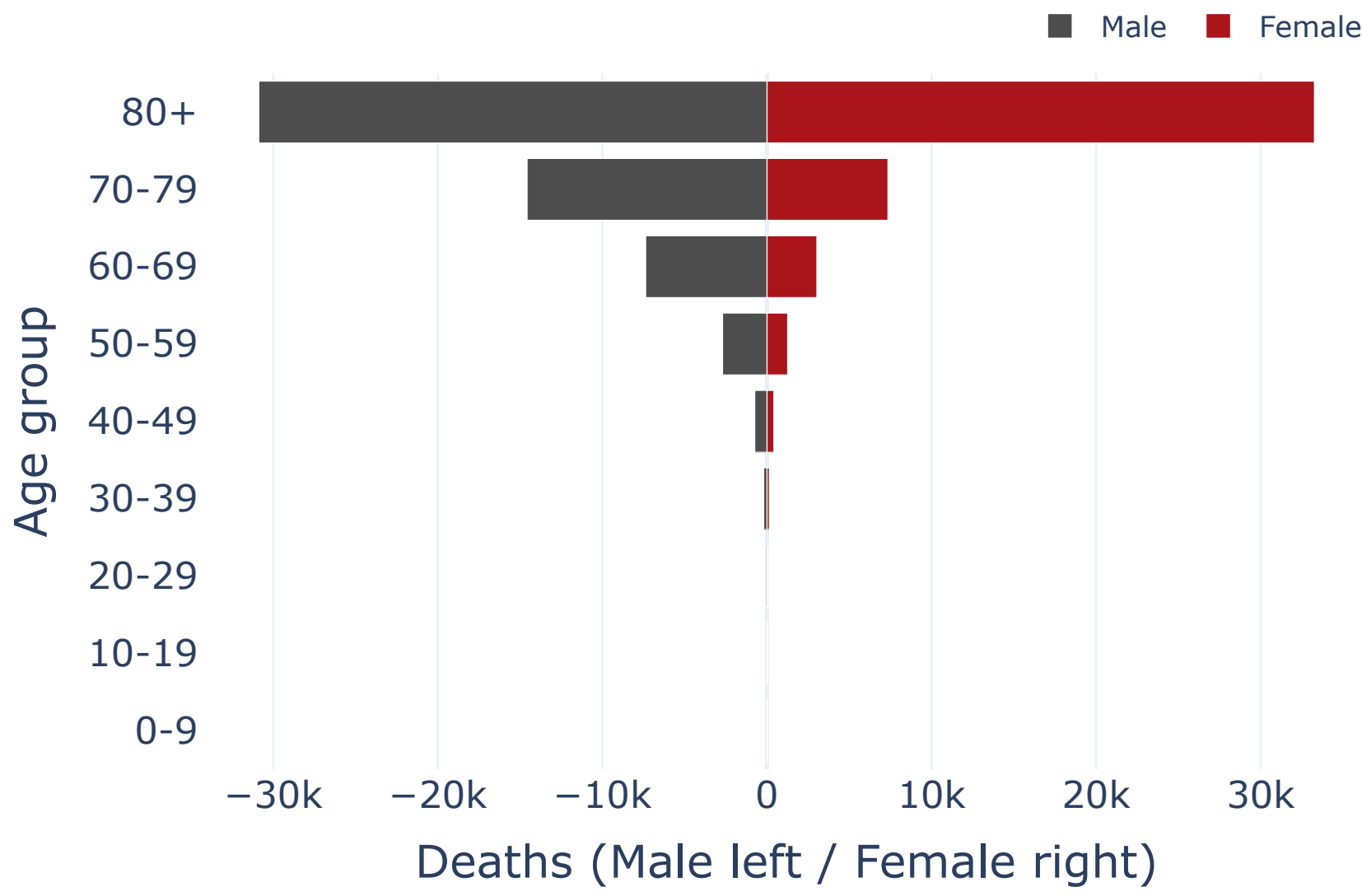


Fig. 4: Age-sex mortality pyramid. COVID-19 deaths are overwhelmingly concentrated in the 70+ age band, with male excess at every age above 50 — a pattern repeated across Europe but especially pronounced in Spain due to a high share of nursing-home residents in early 2020.

Spain’s burden was **not** evenly distributed: by region, by age, or by sex. Public-health planning that treats the country as a single unit misses where the mortality actually fell.

Spain: Total Tourist Nights in All Accommodation (20

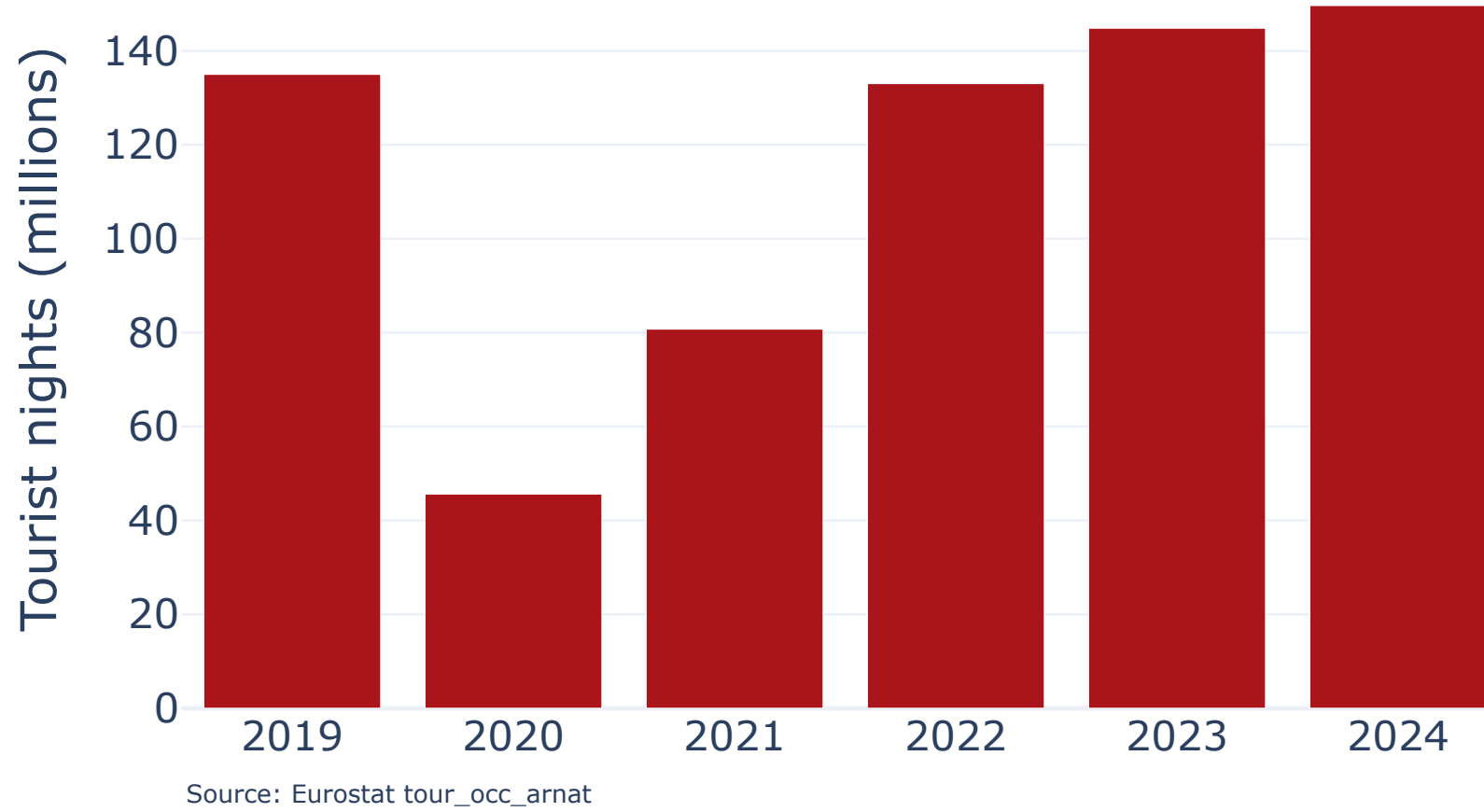


Fig. 5: International tourist arrivals collapsed by 66% in 2020 — the steepest drop on record. Recovery to pre-pandemic levels took until 2023.

Tourism is ~12% of Spanish GDP. The Balearic and Canary Islands saw regional GDP contractions above 20% in 2020 — the largest of any EU NUTS-2 region. This is the channel through which an epidemiological shock became a historic economic shock.

Prophet 2025 Forecast: Sensitivity to Training Windo

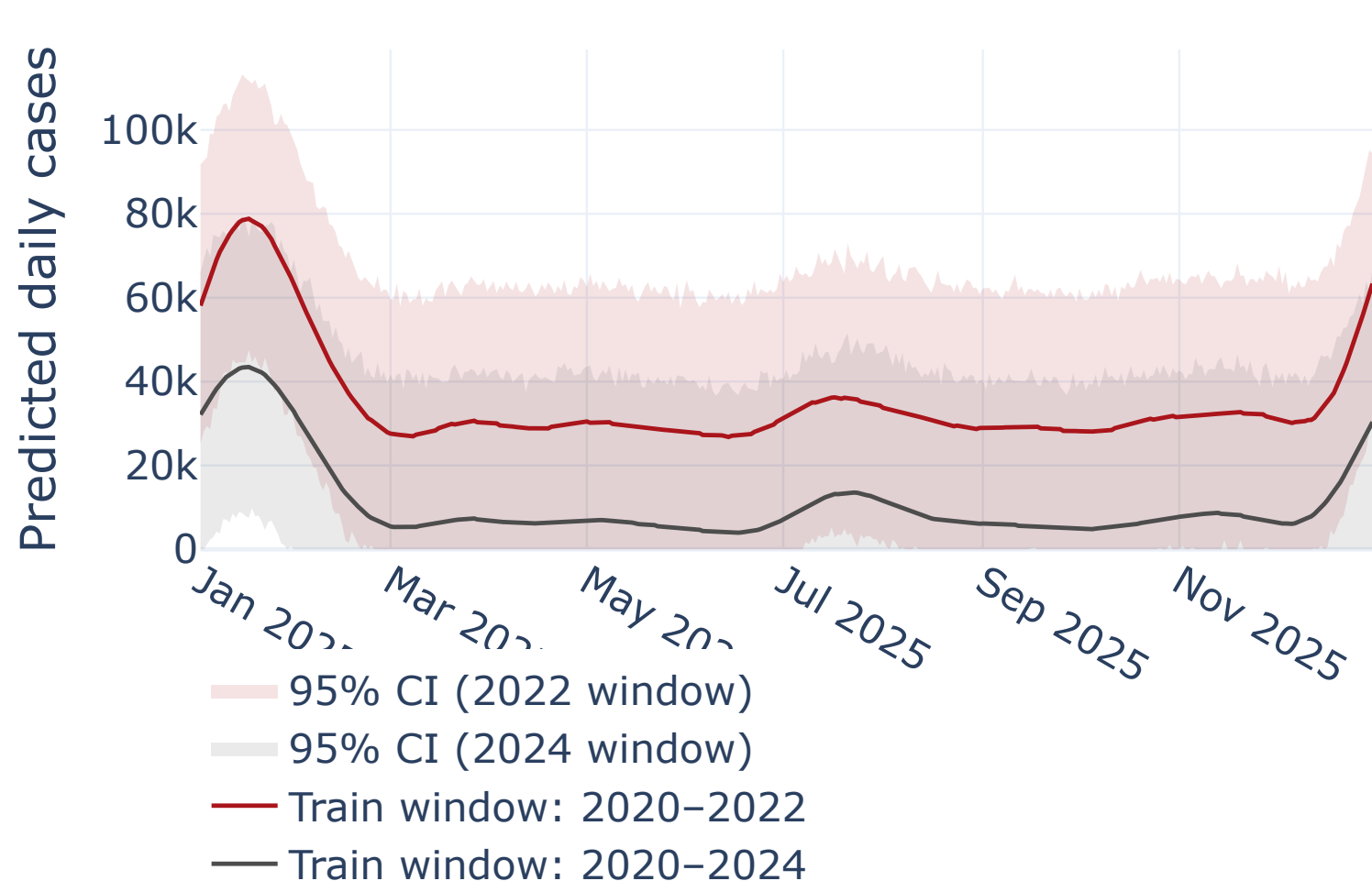


Fig. 6: Prophet forecast of daily cases. *Top*: model trained on 2020–2022 only — it cannot “see” the post-Omicron endemic decline and overshoots dramatically. *Bottom*: the same specification trained on 2020–2024 produces a plausible flat endemic baseline. Same model, very different conclusions. Prophet’s logistic growth floor prevents negative predictions.

Take-away. For a disease in transition from epidemic to endemic, training-window choice dominates model choice. We report both windows deliberately rather than picking the one that “looks right”.

• Spain’s first wave was among the deadliest in Europe; excess mortality shows that confirmed counts **understate** it sharply.

• Mortality is concentrated in the elderly and in inland regions — a national average hides what matters.

• The tourism-dependent economy turned an epidemio-